

Ethical Use of AI in Academic Research

A Practical Guide for Researchers and Doctoral Students

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Purpose of This Document

This document is designed to help researchers and doctoral students make informed, ethical decisions when using artificial intelligence tools in academic research. It is grounded in the content of two workshops on AI in Research and Dissertation Writing conducted by Dr. Sonal R Katyal at the James Lind Institute, Geneva (Session 1: Foundational, 2024; Session 2: Advanced Applications, April 2026).

The guidance here is deliberately practical. It does not take a position against AI tools in research. It takes a position for responsible, transparent, and academically rigorous use of them.

Core Position

AI tools are legitimate research aids when used transparently, with full disclosure, with original intellectual contribution from the researcher, and with all outputs verified against primary sources. Any use that misrepresents AI output as the researcher's own unaided work is a breach of academic integrity.

Section 1: The Three Non-Negotiable Rules

Regardless of the AI tool used, the research context, or institutional policy, three rules apply universally to all AI use in academic research:

Rule 1: DISCLOSE

Document the name, version, and purpose of every AI tool used in your research. This disclosure belongs in your methodology section. Transparency is not optional — it is the foundation of academic trust.

Rule 2: LOG

Maintain an audit trail of significant prompts and AI outputs from the first day of your research. Your research diary or methodology notes are your record of how AI was used. This record is your evidence of responsible use if your integrity is ever questioned.

Rule 3: OWN

Every argument, conclusion, and intellectual claim in your dissertation must be traceable to your own judgment. AI may assist you in finding, organising, or refining information — but the thinking is yours. If you cannot defend an argument in a viva without AI, it does not belong in your dissertation.

Section 2: Categories of AI Use — What Is and Is Not Acceptable

Not all AI use is equivalent. The following framework classifies AI use in academic research into three categories based on ethical risk. Researchers should use this as a self-assessment tool before, during, and after using AI in their work.

GREEN — Ethical AI Use (With Disclosure)

The following uses are considered ethical when properly disclosed in your methodology:

- Using purpose-built screening tools (e.g. Rayyan, Cadima) for abstract screening in systematic reviews — the researcher sets inclusion criteria; the tool executes sorting
- Using statistical analysis tools (e.g. JASP, jamovi) to run analyses on your own original data — these are calculative assistants, not content generators
- Using literature synthesis tools (e.g. Consensus, Elicit, ResearchRabbit) to identify and map existing research — outputs must be verified against primary sources
- Using AI writing tools (e.g. Trinka, Paperpal, Writefull) for grammar, clarity, and academic tone — the intellectual content and argument must remain the researcher's own
- Using AI to explain a concept or methodology that you then verify independently and describe in your own words
- Using AI to structure or outline your argument, which you then develop independently in your own voice

AMBER — Use With Caution (Disclosure Required, Judgment Needed)

The following uses carry ethical risk and require careful disclosure and substantial original contribution:

- Using AI to generate a first draft of a section that you then substantially rewrite — the rewrite must be genuine and traceable to your own thinking
- Using AI to summarise papers — only acceptable if you have read the original papers and verify the AI summary is accurate
- Using AI to suggest research arguments or framings — only acceptable if you have critically evaluated and independently developed those arguments
- Using AI-generated visualisations or tables — only acceptable if underlying data is original, the tool is cited, and outputs are verified

RED — Academic Integrity Violation

The following uses constitute academic misconduct regardless of institutional policy:

- Submitting AI-generated text as your own without disclosure — this is misrepresentation regardless of how the text was produced
- Including AI-generated citations without verifying they exist — AI tools including general-purpose chatbots frequently produce plausible-looking but fabricated references
- Asking AI to write examination answers, assessed coursework, or dissertation chapters that are submitted as your own unaided work
- Sharing confidential research data, participant information, or sensitive institutional data with AI tools — this creates data privacy and ethical risks beyond academic integrity
- Using AI to impersonate your research voice in ways that misrepresent your actual competence to examiners

Section 3: AI Tools by Research Stage

Different stages of the research process have different appropriate tools. The following table maps specialist AI tools to research stages, based on the workshop toolkit developed by Dr. Sonal R Katyal for JLI doctoral and postgraduate researchers.

Important distinction: Specialist tools listed here are purpose-built for academic research and produce citable, reproducible outputs. General-purpose large language models (LLMs) such as ChatGPT or similar tools are not listed here as primary research tools because they do not search real academic databases, cannot provide an audit trail, and frequently produce fabricated citations.

Research Stage	Recommended Tool(s)	Key Benefit
Literature Screening	Rayyan / Cadima (Free)	PRISMA-compliant abstract screening with full audit trail. Researcher sets criteria; tool executes sorting.
Statistical Analysis	JASP / jamovi (Free)	Produces reproducible R-scripts and APA-ready tables. Outputs are citable in methodology.
Citation Mapping	ResearchRabbit (Free)	Maps citation networks from verified academic databases. Does not fabricate references.
Literature Synthesis	Consensus / Elicit (Free Tier)	Draws from real peer-reviewed databases. Outputs show source papers. Verify all claims independently.
Paper Summarisation	Scholarcy (Free Tier)	Turns papers into comparison tables and scaffolds. Use as synthesis starting point, not final output.
Reference Management	Zotero (Free)	Manages citations with full metadata. Use to verify all references before submission.
Academic Writing Quality	Trinka / Paperpal / Writefull (Free Tier)	Trained on peer-reviewed corpora. Enforces academic register,

Integrity Verification	GPTZero / Turnitin AI Detection (Institutional)	hedging, and tense — not just grammar. Checks submissions for AI-generated content before final submission. Use as a self-check tool.
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Critical Note on General-Purpose LLMs

Tools such as ChatGPT, Claude, Gemini, and similar general-purpose AI assistants can support research tasks such as explaining concepts, improving writing clarity, and structuring arguments. However, they must never be used as primary research tools for finding or citing literature. These tools do not search academic databases and regularly produce fabricated citations that appear credible but do not exist. Always verify any source provided by a general-purpose LLM before including it in your work.

Section 4: Precision Prompting for Academic Research

The quality and reliability of AI output in research depends directly on the quality of the input prompt. Vague prompts produce vague — and potentially unreliable — outputs. The following four-part framework was developed for use in academic research contexts.

The 4-Part Precision Prompting Formula

ROLE	Specify who the AI should act as. <i>Example: "Act as an academic peer reviewer specialising in health economics..."</i>
TASK	State precisely what you want the AI to do. <i>Example: "Identify the three main methodological limitations in the following abstract..."</i>
CONTEXT	Provide the relevant background and purpose. <i>Example: "This is for a PhD dissertation on maternal healthcare utilisation in South Asia..."</i>
CONSTRAINT	Set clear boundaries on what the AI should and should not do. <i>Example: "Use only peer-reviewed sources. Do not fabricate citations. Clearly flag any uncertainty."</i>

Why Precision Prompting Matters for Academic Integrity

A vague prompt to a general-purpose AI tool is more likely to produce a vague, generalised, or fabricated response. Adding a CONSTRAINT that explicitly instructs the AI not to fabricate citations and to flag uncertainty significantly reduces — though does not eliminate — the risk of unreliable output. The researcher must still verify all AI output independently.

Section 5: AI Hallucination and Citation Verification

AI hallucination refers to the tendency of large language models to produce confident, plausible-sounding outputs that are factually incorrect or entirely fabricated. This is particularly serious in academic research because hallucinated citations can look entirely credible — with correctly formatted author names, journal titles, volume numbers, and page ranges — but not exist.

How to Identify a Potentially Fabricated Citation

Before including any AI-generated citation in your work, check for the following warning signs:

- The DOI does not resolve to a real paper when searched at doi.org
- The journal name exists but the volume, issue, or page numbers are inconsistent with that journal's publication history
- The author exists but has no record of publishing on that topic
- The paper cannot be found on Google Scholar, Semantic Scholar, PubMed, or SCOPUS
- The paper appears only on AI-generated content aggregator sites, not in academic databases

Verification Protocol — Before Citing Any AI-Generated Reference

1. Search the exact title in Google Scholar (scholar.google.com)
2. Search the DOI at doi.org if one is provided
3. Search the author name combined with the topic in Semantic Scholar or SCOPUS
4. If found, access the actual paper and confirm the claim the AI attributed to it is accurate
5. If not found through any of these methods — do not cite it

Absolute Rule on Citations

Never include a citation in your academic work that you have not personally located, accessed, and read. This applies to all citations regardless of their source. The academic convention that you cite only what you have read is not suspended when AI is used to find sources.

Recommended Tools for Citation Verification

- Google Scholar — scholar.google.com — broad academic search
- Semantic Scholar — semanticscholar.org — AI-assisted but database-grounded
- PubMed — pubmed.ncbi.nlm.nih.gov — for health and biomedical research
- SCOPUS — for indexed journal verification (institutional access required)

- ResearchRabbit — researchrabbit.ai — citation mapping from verified sources
 - Consensus — consensus.app — research synthesis from real peer-reviewed papers
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Section 6: Disclosure Requirements

Academic institutions increasingly require researchers to disclose AI use. However, disclosure requirements vary between institutions, disciplines, and journals. In the absence of a specific institutional policy, the following guidance represents good academic practice.

What to Disclose

- The name and version of each AI tool used (e.g. ChatGPT-4o, Claude 3.5, Trinka, JASP 0.19)
- The purpose for which it was used (e.g. grammar checking, literature screening, data analysis)
- The stage of research at which it was used (e.g. literature review, data analysis, write-up)
- Any limitations of the tool relevant to your research findings

Where to Disclose

- In the Methodology section of your dissertation or paper — this is the standard location
- In a dedicated AI Use Statement if your institution requires one
- In the Acknowledgements section for minor uses such as grammar checking
- In the data availability statement if AI was used in data processing

Sample Disclosure Statement

Example — Methodology Section Disclosure

This research used the following AI tools in the conduct and writing of this dissertation: Rayyan was used for abstract screening during the systematic literature review; the researcher set all inclusion and exclusion criteria independently. Trinka AI was used for grammar and academic tone checking of the final draft; all intellectual content and arguments are the author's own. ChatGPT-4 was used to clarify specific statistical concepts, which were subsequently verified against primary sources. No AI tool was used to generate research findings, conclusions, or cited references.

Section 7: Data Privacy and Confidentiality

When using AI tools in research, data privacy is a distinct and serious concern that sits alongside academic integrity. The following guidance applies specifically to research involving human participants, sensitive institutional data, or confidential information.

What You Must Never Share With AI Tools

- Personal data of research participants — names, contact details, identification numbers, or any data that could identify an individual
- Verbatim interview transcripts or focus group recordings without participant consent for AI processing
- Confidential institutional data, internal reports, or proprietary organisational information
- Unpublished research findings of colleagues shared in confidence
- Patient data or clinical records in health research

Why This Matters

Most commercial AI tools process inputs on external servers. Data shared with these tools may be used to train future models, stored in ways outside your control, or subject to data breaches. Sharing participant data with commercial AI tools without explicit consent and a data processing agreement may violate data protection regulations including GDPR in Europe and equivalent legislation in other jurisdictions.

Practical Rule

Before entering any data into an AI tool, ask: would I be comfortable if this data appeared in a future AI training dataset? If the answer is no — do not enter it. Use anonymised or synthetic data for any AI-assisted analysis.

Section 8: Self-Assessment Checklist

Use the following checklist before submitting any research work in which AI tools were used. A honest self-assessment is the final step in responsible AI use.

#	Before You Submit — Ask Yourself
1	Have I disclosed every AI tool used, its purpose, and the stage at which it was used in my methodology section?
2	Have I personally verified every citation in my reference list — including any suggested by AI tools?
3	Can I defend every argument and conclusion in my work without relying on an AI tool to explain it?
4	Have I maintained an audit trail or research diary documenting significant AI interactions?
5	Have I ensured that no personal data, participant information, or confidential data was shared with any AI tool?
6	Have I read and complied with my institution's current AI use policy?
7	Is the intellectual contribution in this work genuinely mine — or have I allowed AI to substitute for my own thinking?
8	Have I used specialist, citable tools (not just general-purpose chatbots) for methodology-critical tasks such as screening and statistical analysis?

A Note on Intellectual Integrity

The purpose of academic research is to generate original knowledge — to add something to what is already known. AI tools can help researchers work more efficiently, find relevant literature more quickly, and communicate their findings more clearly. They cannot generate original knowledge on a researcher's behalf.

The researcher's intellectual contribution — their judgment about what questions to ask, what evidence to credit, what arguments to make, and what conclusions to draw — is what academic work is ultimately evaluated on. No AI tool can substitute for that contribution, and no disclosure statement can retrospectively legitimise its absence.

Use AI tools deliberately, transparently, and in ways that enhance rather than replace your own thinking. That is the standard this document holds.

Contact & Further Guidance

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